

Passive Component Market (in million dollars)

Types of components	Total market	Indigenous	Imports	Exports
Wound components	1413	424	989	17
Capacitors	269	48	221	40
Resistors	81	5	59	2.83
Piezo-electric crystals	63	—	63	—
Crystal oscillators	54	—	54	—
Total	1880	477	1386	59.83

Source: ELCOMOS report by ELCINA

largely imported from China, Taiwan and South Korea.

Lack of availability of raw material in India is the biggest constraint.

"We are manufacturing wire-wound components like transformers, coils and EMI chokes. But basic copper wire manufacturers are not there in India. That's our main constraint. We have to import the copper wire (magnet wire that goes into the transformer) from Singapore, Korea or China. Then we assemble the product and sell it," says GSP Electronics director Satish Gupta.

Apart from the large small-scale sector that exists, India also has a few very large companies that manufacture wound components in India and also export to the global markets. The key export markets include Singapore, Malaysia, the UK and South America with the total export estimated at \$17 million in FY 2009-10.

Conducive factors. The market is primarily dominated by small-scale indigenous manufacturers. Requirement of lower initial investment and availability of inexpensive labour, qualified professionals and suitable infrastructure facilities are key reasons for the presence of a large number of small-scale indigenous manufacturers in the country.

Crystal oscillators: Negligible indigenous manufacturing

Crystal oscillators are widely used for nearly all electronics products. In some instances, crystal oscillators are used to provide a cost-effective clock signal

for use in a digital or logic circuit. In other instances, they are used to provide an RF signal source. Some of the typical application areas of crystal oscillators include quartz clocks, watches, radio and television transmitters, and mobile phones.

The Indian market for crystal oscillators was estimated at \$54 million in FY 2009-10. All the crystal oscillators used in India are currently imported from Japan and Taiwan. Non-availability of good-quality quartz crystals is one of the biggest reasons why Indian companies have not ventured in this product space. With limited efforts to manufacture the component in India, import will continue to be the major source of crystal oscillators for the Indian electronics industry.

Globally, four of the largest crystal oscillator manufacturers are based in Japan. The other key markets for oscillator manufacturers are Europe and USA. However, in recent times, Chinese and Taiwanese companies have also been successful in designing and manufacturing cost-effective crystals and have started exporting these oscillators to companies located in Europe, India and South America.

Piezoelectric crystals: Story goes the same

Piezoelectric crystals are one of many small-scale energy sources available in the market. Whenever piezoelectric crystals are mechanically deformed or subject to vibration, these generate a small voltage.

The global piezoelectric crystals market was estimated at \$6.91 billion in FY 2011-12. The growth in this market is largely attributed to the emergence of wireless communications and the need for wire-line based data transmission. Apart from this, industrial and consumer applications

are also steering the industry towards a steady growth.

Currently, Japan and South Korea are the two major countries supplying piezoelectric crystal to the world for microprocessor-based devices. China is the largest supplier of piezoelectric crystals for toys and games.

The Indian market for piezoelectric crystals was estimated at \$63 million in FY 2009-10. With no local manufacturing happening, the entire requirement is met through imports.

Way forward

"Looking at the number of manufacturers coming to India, we are very positive that the demand for passive components will increase in India in the near future. We see a huge potential in India. However, there remain several big hurdles to overcome, especially on facilities required for setting up of the passive components manufacturing base in the country," says Mikio Oki, managing director, Murata Electronics (India).

Oki adds, "For the market to be attractive, infrastructure such as electricity, water, road and transportation is going to be one of the key determining factors for manufacturing. The Indian government initiatives to invest and improve in such areas are very important. They will have to take the leadership, in order to make the investment environment attractive enough for manufacturers to consider setting up manufacturing base in India.

"Currently, there is lack of incentives for foreign manufacturers in the industrial zone. In addition to that, the Indian government is relatively slow when it comes to making of policies and setting of directions. As such, we cannot say that India is attractive enough for manufacturing at the present moment. Nonetheless, we are hopeful that the situation will change one day!" ●

The author is executive editor at EFY.

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